

Inhibitory potential of phytochemicals on species-specific breast cancer resistance protein transport activity

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ABSTRACT

The use of herbal alternatives in modern agriculture and healthcare offers a sustainable approach to animal and human health, but raises concerns about safety, particularly regarding phytochemical interactions at membrane transport proteins. Phytochemicals – bioactive compounds found in herbs – can influence the function of the breast cancer resistance protein (BCRP/ABCG2). BCRP is a key efflux transporter in mammals and in particularly abundant in the blood-milk-barrier of lactating animals, with the potential to affect milk quality and safety. This study investigates species-specific interactions of eight selected phytochemicals with BCRP orthologs in humans, cows, sheep and goats, using a transport assay with membrane vesicles derived from species-specific BCRP-overexpressing HEK293 cells. Five phytochemicals, namely apigenin, berberine, kaempferol, N-isobutyldodeca-2E4E8Z10E/Z-tetraenamide and quercetin inhibited BCRP-mediated transport of its prototypic substrate [³H] estrone sulfate by more than 30 % in all species. IC₅₀ values and, assuming competitive inhibition, K_i values were calculated for these compounds. Apigenin and kaempferol were the most potent inhibitors with K_i values <0.1 μM in all species, while N-isobutyldodeca-2E4E8Z10E/Z-tetraenamide was the weakest inhibitor (K_i > 30 μM). No significant interspecies differences in inhibitory potency were observed. Understanding the cross-species effects of bioactive compounds on BCRP activity is essential for predicting their broader biological impacts. Similar inhibition trends across species facilitate interspecies extrapolation and minimize the amount of animal studies needed to further investigate the effect of these phytochemicals on milk composition.

1. Introduction

Modern agriculture faces mounting pressure to use antimicrobials and other veterinary pharmaceuticals responsibly. In response, many farmers are exploring herbal alternatives as a sustainable and holistic approach to animal health (as reviewed by Kuralkar and Kuralkar, 2021).

Herbs are rich in bioactive compounds (phytochemicals) derived from their primary and secondary metabolites. Secondary metabolites such as alkaloids, flavonoids, terpenoids, and phenolic compounds, play vital roles in plant defense and environmental interactions. These compounds have been studied for their antimicrobial, anti-inflammatory, and anticancer properties, making herbs valuable not only as flavourings, but also as natural health-promoting compounds.

However, the use of herbal products is not without risks, as bioactive phytochemicals have the potential to cause drug-phytochemical and nutrient-phytochemical interactions (Stargrove et al., 2007).

The ATP-binding cassette (ABC) efflux transporter, breast cancer resistance protein (BCRP/ABCG2), is highly expressed in organs responsible for absorption, elimination, and distribution and is particularly significant in the mammary glands of lactating humans and dairy animals (Lindner et al., 2013). Research indicates that dietary phytochemicals can interact with BCRP (Zhang et al., 2004; Tan et al., 2013). When consumed by lactating mammals, phytochemicals can transfer into milk, potentially offering health benefits, or enhancing milk's nutritional profile (Valent and Horvath, 2020). However, this transfer could also pose food safety concerns, as phytochemicals interfering with transporter proteins at the blood-milk barrier, have the potential to

Abbreviations: BCRP, breast cancer resistance protein; [³H]E₁S, [³H]estrone sulfate.

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affect the excretion of xenobiotics, such as drugs or toxins, into milk.

There is a significant knowledge gap regarding the effects of increased herbal product use on milk quality and safety, in both lactating women and milk-producing animals. Current data is often species-specific, and little is understood about species-specific differences in BCRP interactions. Species-specific differences in BCRP substrate affinity and efficacy complicate cross-species predictions (Tan et al., 2013). To address this, we have previously developed a screening model using membrane vesicles with BCRP orthologs from humans, cows, goats, and sheep (Le Roux-Pullen et al., 2024). Here we applied this model to investigate the interaction of eight bioactive phytochemicals with species-specific BCRP orthologs. These bioactive phytochemicals were selected based on their well-documented health-promoting effects in both humans and animals, as well as prior evidence of their presence in milk. For some of the compounds there have been evidence of their potential interaction with ABC transporters in a relevant species (see supplementary material Table S01, apigenin, kaempferol, berberine and quercetin for BCRP and anethole and carvacrol for P-gp). Due to financial and practical constraints, the study was limited to eight compounds. The study seeks to advance our understanding of phytochemical-BCRP interactions, contributing to the assessment of the effects of health-promoting phytochemicals, on, among others, milk from lactating mothers and dairy animals.

2. Materials and methods

2.1. Phytochemicals

The phytochemicals used in this study (detailed in the supplementary material Table S01, with information including: a commonly used herb the chosen bioactive phytochemical can be extracted from, the claimed health benefits of the phytochemical in ruminants, the chemical classification of the phytochemical, whether or not it has previously been detected in milk and publications of known BCRP interaction/s where available.) were selected based on complying to the following three criteria:

1. Present in herbs/botanicals that are used in both humans and dairy animals for their health-promoting properties.
2. Known to affect BCRP expression and/or function in one of the listed species, or has been detected in milk.
3. Readily available and soluble in solvents suitable for the chosen *in vitro* assay.

The following pure phytochemical compounds were purchased from MedChemExpress® (Netherlands): allyl methyl sulfide (C₄H₈S, abbreviated in this paper to AMS), anethole (C₁₀H₁₂O), apigenin (C₁₅H₁₀O₅), berberine (C₂₀H₁₈ClNO₄), carvacrol (C₁₀H₁₄O), kaempferol (C₁₅H₁₀O₆) and quercetin (C₁₅H₁₀O₇). N-isobutyldodeca-2E4E8Z10E/Z-tetraenamide (C₁₆H₂₅NO, abbreviated in this paper to 'dodeca') was provided by PhyProof® (Germany). The radio-labelled substrate [³H]estrone sulfate (E₁S) as well as Opti-Fluor® scintillation fluid were purchased from PerkinElmer, Inc. (MA, USA). The MultiScreen HTS glass fibre filter plates and the vacuum manifold filtration devices were purchased from Millipore® (Netherlands). All additional components used for the assay (DMSO, acetonitrile, Tris-HEPES, sucrose, MgCl₂, ATP, and AMP) were purchased from Sigma-Aldrich® (Netherlands).

2.2. Screening for species-specific BCRP-transporter activity inhibition

The transduction of human embryonic kidney (HEK293) cells and isolation of BCRP-overexpressing membrane vesicles was done as previously described (Le Roux-Pullen et al., 2024). BCRP-sequences from 4 different species were used, namely human, bovine, caprine and ovine. The functional activity of BCRP with respect to the chosen model substrate, tritium-labelled estrogen sulfate [³H]E₁S, was previously

published by us (Le Roux-Pullen et al., 2024).

These species-specific vesicles were used to determine the inhibitory potential of eight phytochemicals (at a screening concentration of 200 μM) on the transport of [³H]E₁S. Experiments were performed in triplicate on three different days.

A vesicular transport inhibition assay with rapid filtration technique was used, similarly to published work for vesicles expressing human ABC-transporters (Lempers et al., 2016). Briefly, the phytochemical agents were dissolved to a concentration of 20 mM in dimethyl sulfoxide (DMSO), with the exception of dodeca, which was dissolved in acetonitrile (CH₃CN). These stock solutions were then further diluted with the assay buffer (10 mM Tris-HEPES and 250 mM sucrose, pH 7.4), to the desired test concentration of 200 μM. For incubation with the individual phytochemicals, a total volume of 30 μL with 7.5 μg species-specific vesicles in assay buffer, supplemented with 10 mM MgCl₂ and either 4 mM ATP, or 4 mM AMP (serving as a negative control) and 0.075 uCi [³H]E₁S, were added to each well of a 96-well plate and incubated for 1 min at 37 °C. Incubation was stopped by placing the plate on ice and by adding 150 μL ice cold stop buffer. The vesicle reaction mixtures were transferred onto Millipore MultiScreen HTS 96-wells filter plates that had been preincubated with TS buffer. To separate the vesicles from the incubation solution, the samples were filtered using a MultiScreen HTS-vacuum manifold filtration device, after which they were washed twice with 200 μL TS buffer. The washed filters were punched out in scintillation counting vials. After adding 2 mL of Opti-Fluor scintillation fluid, the samples were counted using a liquid scintillation counter (Hidex 600 SL).

An average was calculated for three data points per phytochemical-transporter interaction and net final values were calculated subtracting the ATP values by the AMP values. Negative calculated values were set to 0. The transport of the model substrate [³H]E₁S was expressed as percentage transported compared to 100 % transport in the control samples (no phytochemical added, only incubated in 1 % DMSO or CH₃CN solvent). For each phytochemical the screening assay was performed in triplicate on three different days. The [³H]E₁S transport values were presented as mean percentage plus standard deviation (STDEV). Phytochemicals inhibiting transport of [³H]E₁S for more than 30 % at 200 μM were considered potential inhibitors of interest.

2.3. Concentration-dependent inhibition of species-specific transporter activity

The vesicular transport assay with rapid filtration technique detailed above, was also used to further determine concentration-dependent inhibition and assess 50 % inhibitory concentrations (IC₅₀) of each of the phytochemicals that inhibited the ATP-dependent BCRP-transport of [³H]E₁S for more than 30 % at the screening concentration of 200 μM. Transport studies were performed as described above, using the following concentration ranges: 0.003, 0.03, 0.1, 0.3, 1, 3 and 30 μM for apigenin and kaempferol, 0.75, 1.5, 7.5, 37.5, 75, 150 and 300 μM for berberine, 1, 5, 10, 25, 50, 100 and 200 μM for dodeca and 0.1, 0.3, 1, 3, 10, 30 and 100 μM for quercetin. The concentration ranges used per phytochemical were determined through pre-trial experiments, taking the solubility of each compound into consideration. Each concentration was tested in triplicate on three separate days.

The inhibition of ATP-dependent uptake of [³H]E₁S was expressed as a percentage of the control and summarized as mean ± STDEV.

Curve fitting of the concentration-dependent inhibition of transport and subsequent calculation of IC₅₀ values was established by nonlinear regression analysis (with the top constrained to 100 % transport and the bottom to 0 %) using GraphPad Prism version 9.3.1 (GraphPad Software Inc., CA, USA). Species-specific IC₅₀ values were compared for each phytochemical using a two-way ANOVA, followed by Tukey's multiple comparison test as recommended by GraphPad Prism version 10 for pairwise comparisons (comparing columns within each row). The simultaneous statistical testing reduces the false discovery rate, giving

more certainly to our findings.

Subsequently, the inhibition constant (K_i) values were calculated from the best-fitting IC_{50} values according to the Cheng-Prusoff equation, assuming that inhibition is competitive and that only one inhibitor-binding site is involved, according to the following equation:

$$K_i = IC_{50} / (1 + [S]/K_m)$$

where $[S]$ is the concentration of $[^3H]E_1S$ used during the experiments (4.5 nM) and K_m is the Michaelis constant. The K_m values for $[^3H]E_1S$ in the various species-specific BCRP-orthologs have previously been published (Le Roux-Pullen et al., 2024).

3. Results

The study resulted in two complementary datasets: the initial screening for species-specific inhibition of BCRP transporter activity by selected phytochemicals and the subsequent evaluation of the potential inhibitors' concentration-dependent inhibitory effects, which enabled the calculation of their inhibition affinity parameters (IC_{50} and K_i values).

3.1. Screening for species-specific BCRP-transporter activity inhibition

Fig. 1 shows the inhibitory potency of the phytochemicals against the ATP-dependent uptake of $[^3H]E_1S$, in four species. For all compounds the same trend was seen across the species investigated (i.e. each compound tested was either a potential inhibitor of BCRP in all four species, or not). Five phytochemicals inhibited $[^3H]E_1S$ transport at 200 μM for more than 30 % in all species, namely the flavonoids apigenin, kaempferol and quercetin, the isoquinolone alkaloid berberine and the alkamide dodeca. Transport was inhibited for less than 30 % by AMS, anethole and carvacrol and therefore these phytochemicals were not considered for further IC_{50} assessment. Finally, for the latter phytochemicals, in some cases modest stimulation of transport was observed.

3.2. Concentration-dependent inhibitory effects

The ATP-dependent BCRP-mediated transport of $[^3H]E_1S$ after exposure to increasing concentrations of apigenin, berberine, dodeca, kaempferol and quercetin was determined in four species, and is presented in Fig. 2. The estimated IC_{50} values from the fitted lines are summarized in Table 1 and the calculated K_i values in Table 2.

P -values ranged between 0.217 (human *versus* ovine for dodeca) and > 0.999. There were no statistically significant differences for any of the phytochemicals when comparing between species (alpha 0.05, see Table S02 supplementary material).

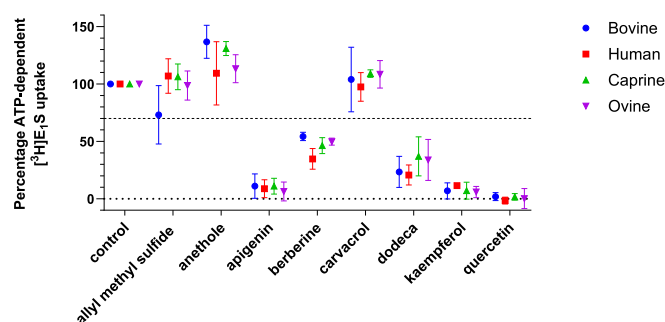


Fig. 1. The inhibitory potency of selected phytochemicals (200 μM) on the ATP-dependent uptake of $[^3H]E_1S$ into membrane vesicles overexpressing species-specific BCRP. Transport is expressed in % of the solvent controls and presented as mean $\pm$ STDEV. The grey reference line indicates 70 % transport. Inhibition of more than 30 % (i.e. compounds below the reference line) was considered relevant to investigate further. Experiments were performed in triplicate.

Overall apigenin and kaempferol were the strongest inhibitors with the highest affinity for BCRP, with IC_{50} values of less than 0.3 μM in all species and K_i values ranging between 0.02 and 0.1 μM . Dodeca was the weakest inhibitor of $[^3H]E_1S$ transport of these 5 phytochemicals, with IC_{50} values above 25 μM and K_i values ranging between 31 and 220 μM , but with large variability between replicates.

4. Discussion

This study revealed consistent trends in the inhibition of the Breast Cancer Resistance Protein (BCRP) across human, bovine, caprine, and ovine orthologs by selected bioactive phytochemicals. These findings further highlight the evolutionary conservation of BCRP function, as discussed in the review by Sarkadi et al. (2006). Eight bioactive phytochemicals with known health-promoting properties (a summary of the compounds can be found in the supplementary materials in Table S01) were assessed for their inhibitory effects on BCRP-mediated transport of $[^3H]E_1S$, using membrane vesicles derived from species-specific BCRP-overexpressing HEK293 cells. Apigenin, berberine, dodeca, kaempferol, and quercetin reduced BCRP-mediated transport by more than 30 % in all species during screening. Subsequently, when comparing concentration-dependent inhibition, apigenin and kaempferol showed the strongest inhibitory effects across all species, followed by quercetin and berberine.

AMS, anethole, and carvacrol exhibited negligible inhibition. Anethole and carvacrol have both been previously reported to inhibit P-glycoprotein (P-gp/ABCB1). Although there are some overlap in substrate and inhibitor specificity between P-gp and BCRP, BCRP possesses distinct structural and functional features that result in notable differences in substrate recognition and transport mechanisms. It is well established that the physicochemical properties of a compound – such as planarity, molecular size, and hydrogen bonding potential – are critical determinants of its interaction with membrane transporters (Mao and Unadkat, 2015). AMS, a small and volatile metabolite derived from garlic, is not known to inhibit BCRP. In contrast, aged garlic extract (AGE) has demonstrated BCRP inhibitory activity, suggesting that other, more structurally complex and polar organosulfur compounds, such as S-allyl cysteine, are likely responsible for this effect. In some cases, these compounds demonstrated modest stimulation of BCRP-mediated transport, suggesting a dual modulatory role, dependent on concentration and conditions. Such stimulatory effects could result from allosteric interactions, more often observed in *in vitro* assays of ABC transporter activity (Giri et al., 2009; Irobalieva et al., 2023).

Despite their differing biological activities, the physicochemical profiles of anethole, carvacrol, and AMS, characterized by low molecular weight and high lipophilicity (LogP values of approximately 3.4, 2.9, and 1.8, respectively), strongly support passive diffusion as the primary mechanism underlying their transfer into milk.

To compare the IC_{50} values between species, the $\log(IC_{50})$ values of 3 independent experiments, per phytochemical per species, were used and analysed together, instead of conducting multiple independent *t*-tests, assuring that the Type I error rate was not inflated. The Tukey's multiple comparisons test uses the pooled variance from the ANOVA, further ensuring consistent and reliable estimates of variability across all the comparisons. The inhibitory trends were consistent across species, with no statistically significant differences in the $\log(IC_{50})$ values of each tested compound. This interspecies uniformity underscores the evolutionary conservation of BCRP and suggests that these phytochemicals could serve as reliable inhibitors in both human and veterinary contexts.

A previous study has identified apigenin, berberine, kaempferol, and quercetin as BCRP inhibitors in humans (Tan et al., 2013), but ours is the first study confirming their cross-species inhibitory potential. In our study apigenin and kaempferol were overall the strongest inhibitors, followed by quercetin. This is consistent with an inhibitory study done in Sf9/hBCRP vesicles using Ko143 as model substrate (Tan et al., 2013).

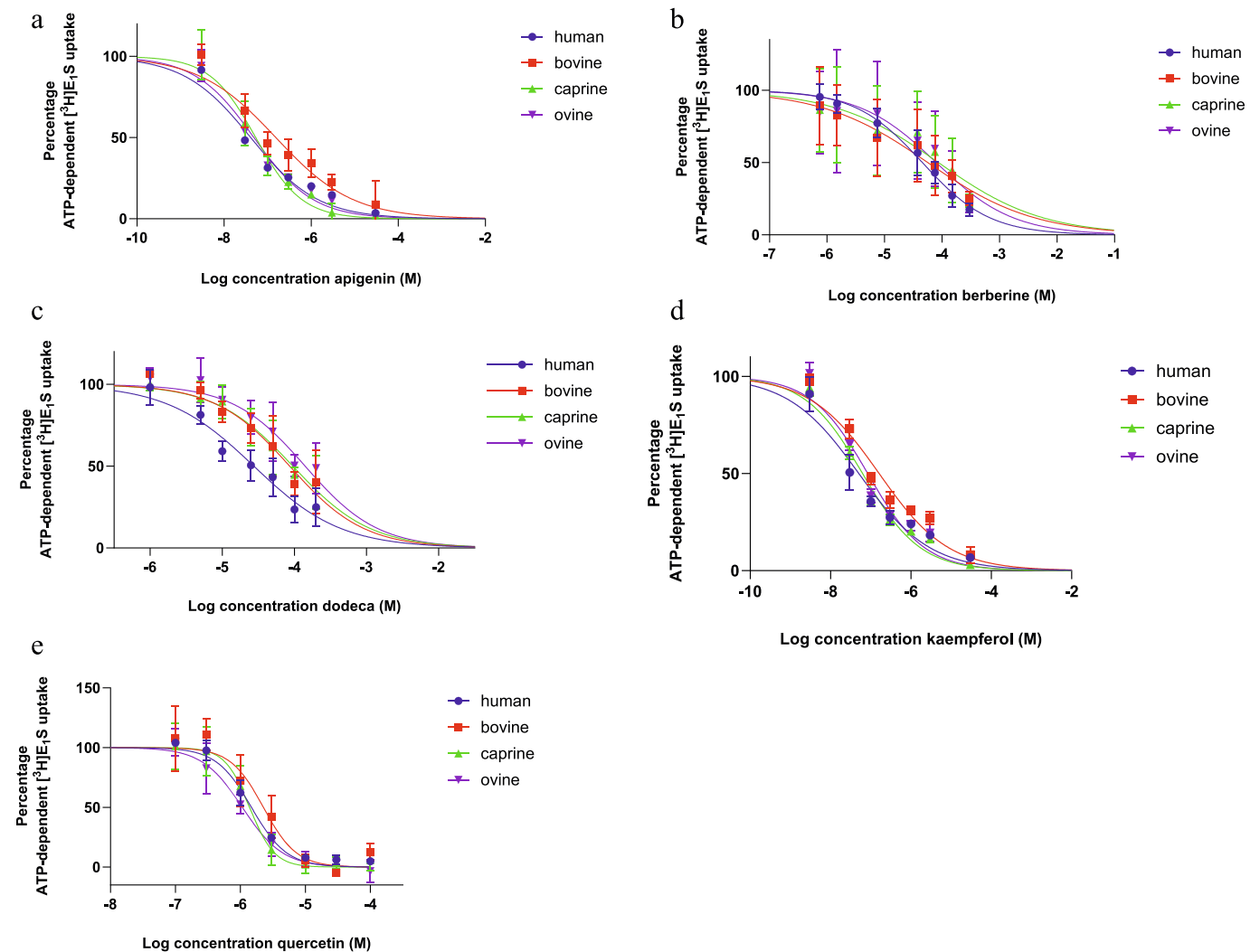


Fig. 2. a to e. The ATP-dependant BCRP-mediated transport of [³H]E₁S in four different species, after exposure to increasing concentrations of the phytochemicals apigenin (a), berberine (b), dodeca (c), kaempferol (d) and quercetin (e). Experiments were performed in triplicate. Data is presented as mean ± STDEV.

Table 1

The best fit IC₅₀ value (μM) for each of the five phytochemicals investigated (apigenin, berberine, dodeca, kaempferol, quercetin) in four species, calculated using a non-linear regression model.

	Best fit IC ₅₀ value Human	Best fit IC ₅₀ value Bovine	Best fit IC ₅₀ value Caprine	Best fit IC ₅₀ value Ovine
apigenin	0.04	0.15	0.05	0.05
berberine	45.71	56.23	87.1	79.43
dodeca	27.54	85.11	95.50	141.25
kaempferol	0.05	0.16	0.06	0.08
quercetin	1.48	2.24	1.38	1.07

Dodeca (N-isobutyl-dodeca-2E4E8Z10E/Z-tetraenamide), a bioactive tetraene found in *Echinacea* preparations (Wölkart et al., 2009) and the major constituent of *Echinacea purpurea* in human plasma, showed weak inhibition in our study, aligning with prior findings of limited activity at clinically relevant concentrations (Awortwe et al., 2021). Especially for this phytochemical, there was a large variation among the data points. The reference substance used in this study consist of a mixture of two isomers and is fairly instability, which complicate *in vitro* assessments.

Previous research has demonstrated that dietary flavonoids can impact drug absorption and distribution (Pulido et al., 2006; Song et al., 2013; Fan et al., 2019). BCRP plays a critical role in the excretion of xenobiotics and nutrients, including their secretion into milk. Inhibition

Table 2

The Ki values (in μM) for each of the five phytochemicals investigated (apigenin, berberine, dodeca, kaempferol, quercetin) in four species, calculated using the Cheng–Prusoff equation with the best fitting IC₅₀ value. Data is presented as mean ± STDEV.

	Ki value Human	Ki value Bovine	Ki value Caprine	Ki value Ovine
apigenin	0.023 ± 0.001	0.1 ± 0.1	0.03 ± 0.02	0.03 ± 0.01
berberine	28.2 ± 17.5	51.9 ± 54.3	48.33 ± 54	64.7 ± 70.9
dodeca	30.6 ± 14.1	110.9 ± 78.4	121.6 ± 79.5	219.2 ± 193.3
kaempferol	0.066 ± 0.004	0.19 ± 0.03	0.059 ± 0.004	0.088 ± 0.002
quercetin	1.5 ± 0.3	2.3 ± 1.2	1.4 ± 0.6	1.1 ± 0.4

of BCRP could significantly alter milk composition, impacting its safety and nutritional value, especially in food-producing animals. Understanding the cross-species effects of bioactive compounds on BCRP activity is essential for predicting their broader biological impacts, including drug interactions and nutrient bioavailability. BCRP inhibition by flavonoids such as apigenin, kaempferol, and quercetin could therefor influence the excretion of BCRP substrates into milk, potentially altering its safety and quality. This is particularly relevant in food-producing animals, where the excretion of veterinary drugs and other

xenobiotics into milk must be closely monitored. Similar inhibition trends for BCRP across species, as seen in our study, facilitate inter-species extrapolation and minimize the amount of animal studies needed to further investigate the inhibitory effect of these phytochemicals.

Phytochemicals have previously been shown to induce BCRP expression (Ebert et al., 2007). The membrane vesicle model cannot evaluate cellular transporter regulation. Not only the binding affinity, but also the abundance of a transporter in individual tissue, accounts for the overall impact of the transporter (Haritova et al., 2006). This cumulative influence ultimately determines the biological significance of ABC transporters and their interaction with compounds. This is an area that should be further investigated to fully assess the potential impact of the investigated phytochemicals on food safety.

In our study competitive inhibition is assumed as the key inhibitory mechanism and the well-established Michaelis-Menten equation used to interpret the effects of the inhibitor on the transporter activity. While this study establishes cross-species trends in BCRP inhibition, future research should elucidate the precise mechanisms of inhibition and determine whether the same trends are seen for the compounds that are also BCRP substrates. The latter would require transport studies quantifying intracellular phytochemical concentrations, which is analytically demanding. With the higher throughput of the inhibition assay, a larger number of compounds can however be evaluated.

In our study apigenin, kaempferol and quercetin showed inhibitory effects at low concentrations. Specific data on blood concentration levels of phytochemicals in ruminant species such as cows, goats, or sheep are scarce. Most available studies focus on the presence and effects of these flavonoids in humans. Reported mean plasma concentrations of apigenin, kaempferol and quercetin in humans are 10.6 nmol/L, 57.9 nmol/L and 80.2 nmol/L, respectively, following dietary intake (Cao et al., 2010). A study with quercetin in bovines showed a comparable blood concentration of 90 nmol/L (Berger et al., 2012). Comparing *in vivo* plasma concentrations of phytochemicals to their *in vitro* K_i values is a widely accepted approach for assessing the potential clinical relevance of transporter inhibition, as outlined in the EMA drug-drug interaction (DDI) guidelines. Using the $[I]/K_i$ ratio - where $[I]$ represents the inhibitor's plasma concentration - we calculated values of 0.46 for apigenin, 0.88 for kaempferol, and 0.053 for quercetin, with respect to human BCRP. Applying the commonly used threshold of $[I]/K_i \geq 0.1$ for potential clinical relevance, these results suggest that apigenin and kaempferol may reach systemic concentrations sufficient to inhibit BCRP under physiological conditions. It should be noted, however, that these values are based on mean systemic plasma concentrations, and local concentrations may be higher and thus more relevant for transporter inhibition. Further *in vivo* studies or physiologically based pharmacokinetic (PBPK) modelling would be required to confirm these inhibitory effects in a clinical context. The bioavailability and metabolism of these compounds can vary significantly between species due to differences in digestive physiology and metabolism, therefore, extrapolating human data to ruminants may not provide accurate information.

5. Conclusion

To our knowledge, our study is the first to investigate potential species-differences in the inhibitory potency of (phyto)chemicals on BCRP transport in lactating mammals, by using a fast throughput membrane vesicle assay using species-specific BCRP.

In summary, the analysis of the inhibitory potency of BCRP inhibition by five phytochemicals -apigenin, berberine, dodeca, kaempferol, and quercetin - showed consistent inhibitory trends across the species included in our study, inhibiting the transport of the high affinity BCRP-substrate $[^3H]E_1S$ similarly between the four species investigated. The study further highlighted differences in the strength of inhibition among these compounds, in a manner similar to results from previous studies.

The species-specific membrane vesicular transport assay used in our study provides a useful risk assessment tool for BCRP-interaction as it

offers a reliable method to evaluate if a compound has the potential to affect transport across biological barriers, such as the blood-milk barrier, ultimately affecting the composition and safety of milk.

Our data can serve as input for physiologically based pharmacokinetic (PBPK) modelling. To accurately predict BCRP kinetics and its impact on food safety, comprehensive models incorporating more data on transporter expression, binding affinities, and tissue distribution, are needed.

Research data

The data that support the findings of this study are available from the corresponding author, L. Le Roux-Pullen upon reasonable request.

CRediT authorship contribution statement

Lérica le Roux-Pullen: Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis, Conceptualization. **Jeroen J.M.W. van den Heuvel:** Writing – review & editing, Validation, Methodology, Investigation, Conceptualization. **Ilse R. Dubbelboer:** Writing – review & editing, Visualization, Formal analysis, Data curation. **Frans G.M. Russel:** Writing – review & editing. **Ronette Gehring:** Writing – review & editing, Supervision, Conceptualization. **Jan B. Koenderink:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.tiv.2025.106128>.

Data availability

Data will be made available on request.

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